

Education

- 2024– Present **Indian Institute of Science, Bangalore**
M.TECH (RESEARCH) COMPUTER SCIENCE
GPA: **9.7/10**
Courses: Design and Analysis of Algorithms, Randomized Algorithms, Topics in Geometric Algorithms
- 2019–2023 **Indian Institute of Technology Delhi**
B.TECH ELECTRICAL ENGINEERING
GPA: **9.381/10**
Minor in Artificial Intelligence
Courses: Reinforcement Learning and Stochastic Control; Online Learning, Bandits and Optimization; Convex Optimization; Concentration Inequalities; Machine Learning; Probability and Stochastic Processes; Mathematics for Machine Learning

Experience

- Aug 2024– Present **Indian Institute of Science, Bangalore**
RESEARCH STUDENT
Advisor - Prof. Siddharth Barman
Working on problems in Learning Theory, Game Theory and Approximation Algorithms.
Teaching assistant for the graduate course Design and Analysis of Algorithms.
- Aug 2023– June 2024 **Microsoft Research, Bangalore**
RESEARCH INTERN
Manager - Dr. Gaurav Sinha
Worked on new bandit models better suited to industrial application in advertising and developed algorithms with theoretical guarantees and empirical performance.
- 2019–2023 **Indian Institute of Technology Delhi**
UNDERGRADUATE
Advisor - Prof. Arpan Chattopadhyay
Undergraduate Thesis [Part 1](#) - Inverse Reinforcement Learning with Constraint Recovery and [Part 2](#) - Linear Rotting Bandits [won **Best Undergraduate Thesis Award**]
Teaching assistant for the graduate course on Reinforcement Learning and Stochastic Control.
- Summer 2022 **Mastercard AI Garage, New Delhi**
ARTIFICIAL INTELLIGENCE INTERN
Built a semi-supervised deep learning pipeline for fraud detection in credit card transaction.

Publications

- 1 Siddharth Barman, [Nirjhar Das](#), Shivam Gupta, Kiran Shiragur (Alphabetical Order). “**Welfarist Formulations for Diverse Similarity Search**”. *Accepted* in The Fourteenth International Conference on Learning Representations.
[Paper](#) (https://www.arxiv.org/abs/2602.08742), [ICLR 2026](#)
- 2 Nirjhar Das, Mohit Sharma, Praharsh Nanavati, Kiran Shiragur, Amit Deshpande. “**Cost Efficient Fairness Audit Under Partial Feedback**”. *Accepted* in Reliable ML Workshop at The Thirty-Ninth Annual Conference on Neural Information Processing Systems
[Paper](#) (https://www.arxiv.org/abs/2510.03734), [Reliable ML Workshop, NeurIPS 2025](#)

- 3 Nirjhar Das, Souradip Chakraborty, Aldo Pacchiano, Sayak Ray Chowdhury. “**Active Preference Optimization for Sample Efficient RLHF**”. *Accepted* in European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases 2025
Paper ↗ (<https://arxiv.org/abs/2402.10500>), **ECML PKDD 2025**
- 4 Ayush Sawarni, Nirjhar Das, Siddharth Barman, Gaurav Sinha. “**Generalized Linear Bandits with Limited Adaptivity**”. *Accepted* as **Spotlight** in The Thirty-eighth Annual Conference on Neural Information Processing Systems
Paper ↗ (<https://arxiv.org/abs/2404.06831>), **NeurIPS 2024 Spotlight**
- 5 Nirjhar Das, Gaurav Sinha. “**Linear Contextual Bandits with Hybrid Payoff: Revisited**”. *Accepted* in European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases 2024
Paper ↗ (<https://arxiv.org/abs/2406.10131>), **ECML PKDD 2024**
- 6 Nirjhar Das, Arpan Chattopadhyay. “**Inverse Reinforcement Learning With Constraint Recovery**”. **Best Paper Award** in 10th International Conference on Pattern Recognition and Machine Intelligence
Paper ↗ (<https://arxiv.org/abs/2305.08130>), **PReMI 2023**
- 7 Mustafa Chasmai, Nirjhar Das, Aman Bhardwaj, Rahul Garg. “**A View Independent Classification Framework for Yoga Postures**”. Springer Nature Computer Science, Vol 3.
Paper ↗ (<https://arxiv.org/abs/2206.13577>), **SNCS Sept 2022**

Honors and Awards

- 2025 **Reliance Foundation Postgraduate Scholarship:** Received the scholarship awarded to top 100 students out of 10,000+ applicants across India.
- 2024 **Adobe-IISc Research Workshop:** Won the *1st Runner Up* award in Student Research Presentation in IISc Bangalore sponsored by Adobe Research India.
- 2024 **Microsoft Travel Grant:** Awarded for presenting our work in ECML PKDD 2024.
- 2023 **Best Paper Award:** Won the award at International Conference on Pattern Recognition and Machine Intelligence (PReMI) 2023.
- 2023 **Best Undergraduate Thesis Award:** Won the award among 160+ students in the EE department for research carried out in BTech Project.
- 2023 **Amazon ML Challenge 2023:** Achieved *6th* rank out of 5000+ teams by building a deep-learning model for the prediction of package length from item description.
- 2019–2023 **Top 7% of the Batch:** Awarded in a total of 4 semesters out of 8 during undergraduate.
- 2019 **Indian School Certificate:** Achieved *2nd* rank all over India out of 100,000+ students in Class XII Board (High School Graduation) Exam with 499/500 marks.
- 2019 **Swami Vivekananda Scholarship:** Awarded by *Hon'ble Chief Minister* of West Bengal for excellent performance in Class XII Board (High School Graduation) Exam.

Skills

Python, C, C++, Java

Programming Languages

NumPy, SciPy, Matplotlib, Scikit Learn, Pandas, Cvxpy

Data Science Libraries

Academic Service and Extracurricular Activities

- **Reviewer** for NeurIPS 2025, IoT Journal 2024, FSTTCS 2025
- **Organizer** of [Bangalore CS Theory Seminar](https://www.csa.iisc.ac.in/theoryseminars/) (<https://www.csa.iisc.ac.in/theoryseminars/>)